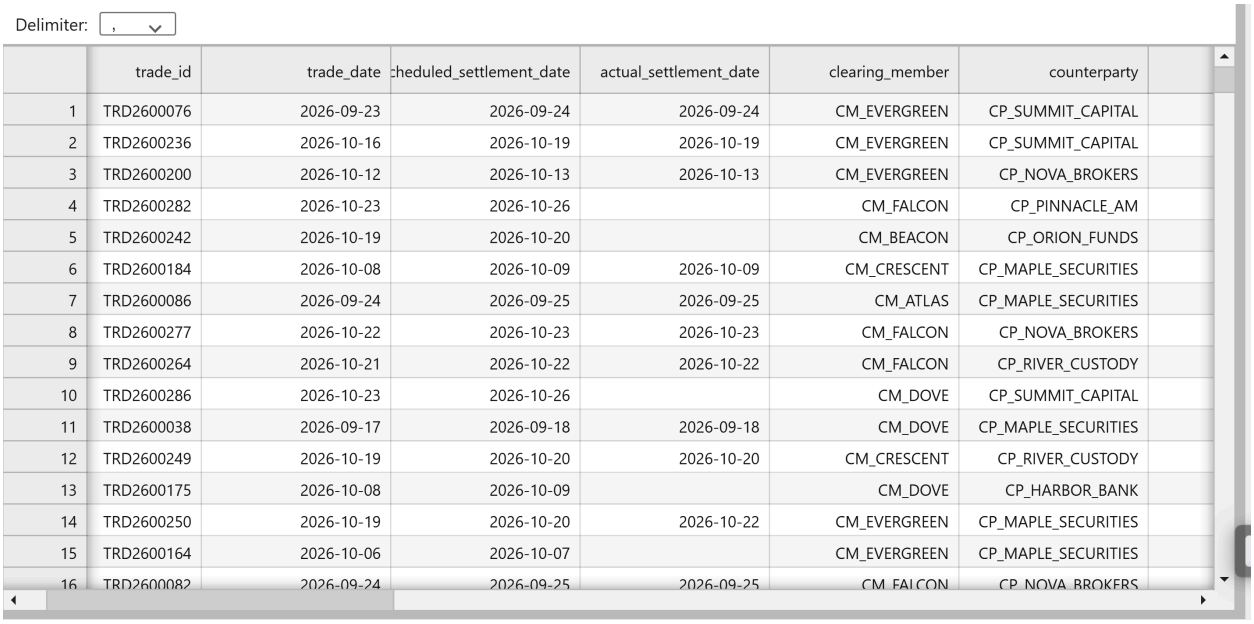
SQL Queries on Clearing Risk Dataset

Cleaning the Erroneous Dataset:



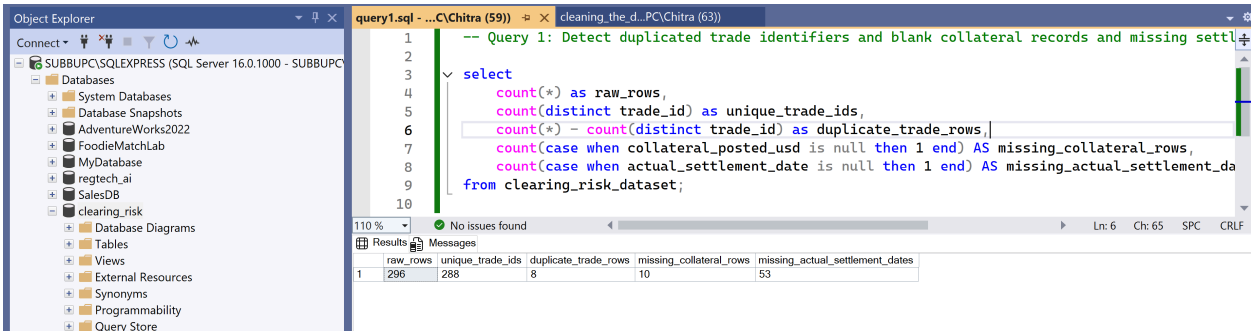
Delimiter: ,

	trade_id	trade_date	cheduled_settlement_date	actual_settlement_date	clearing_member	counterparty
1	TRD2600076	2026-09-23	2026-09-24	2026-09-24	CM_EVERGREEN	CP_SUMMIT_CAPITAL
2	TRD2600236	2026-10-16	2026-10-19	2026-10-19	CM_EVERGREEN	CP_SUMMIT_CAPITAL
3	TRD2600200	2026-10-12	2026-10-13	2026-10-13	CM_EVERGREEN	CP_NOVA_BROKERS
4	TRD2600282	2026-10-23	2026-10-26		CM_FALCON	CP_PINNACLE_AM
5	TRD2600242	2026-10-19	2026-10-20		CM_BEACON	CP_ORION_FUNDS
6	TRD2600184	2026-10-08	2026-10-09	2026-10-09	CM_CRESCENT	CP_MAPLE_SECURITIES
7	TRD2600086	2026-09-24	2026-09-25	2026-09-25	CM_ATLAS	CP_MAPLE_SECURITIES
8	TRD2600277	2026-10-22	2026-10-23	2026-10-23	CM_FALCON	CP_NOVA_BROKERS
9	TRD2600264	2026-10-21	2026-10-22	2026-10-22	CM_FALCON	CP_RIVER_CUSTODY
10	TRD2600286	2026-10-23	2026-10-26		CM_DOVE	CP_SUMMIT_CAPITAL
11	TRD2600038	2026-09-17	2026-09-18	2026-09-18	CM_DOVE	CP_MAPLE_SECURITIES
12	TRD2600249	2026-10-19	2026-10-20	2026-10-20	CM_CRESCENT	CP_RIVER_CUSTODY
13	TRD2600175	2026-10-08	2026-10-09		CM_DOVE	CP_HARBOR_BANK
14	TRD2600250	2026-10-19	2026-10-20	2026-10-22	CM_EVERGREEN	CP_MAPLE_SECURITIES
15	TRD2600164	2026-10-06	2026-10-07		CM_EVERGREEN	CP_MAPLE_SECURITIES
16	TRD2600082	2026-09-24	2026-09-25	2026-09-25	CM_FALCON	CP_NOVA_BROKERS

```
CREATE VIEW clearing_risk_clean AS
WITH RankedTrades AS (
SELECT *,
ROW_NUMBER() OVER (PARTITION BY trade_id ORDER BY trade_id) AS rn
FROM clearing_risk_dataset
)
SELECT *
FROM RankedTrades
WHERE rn = 1;
GO
```

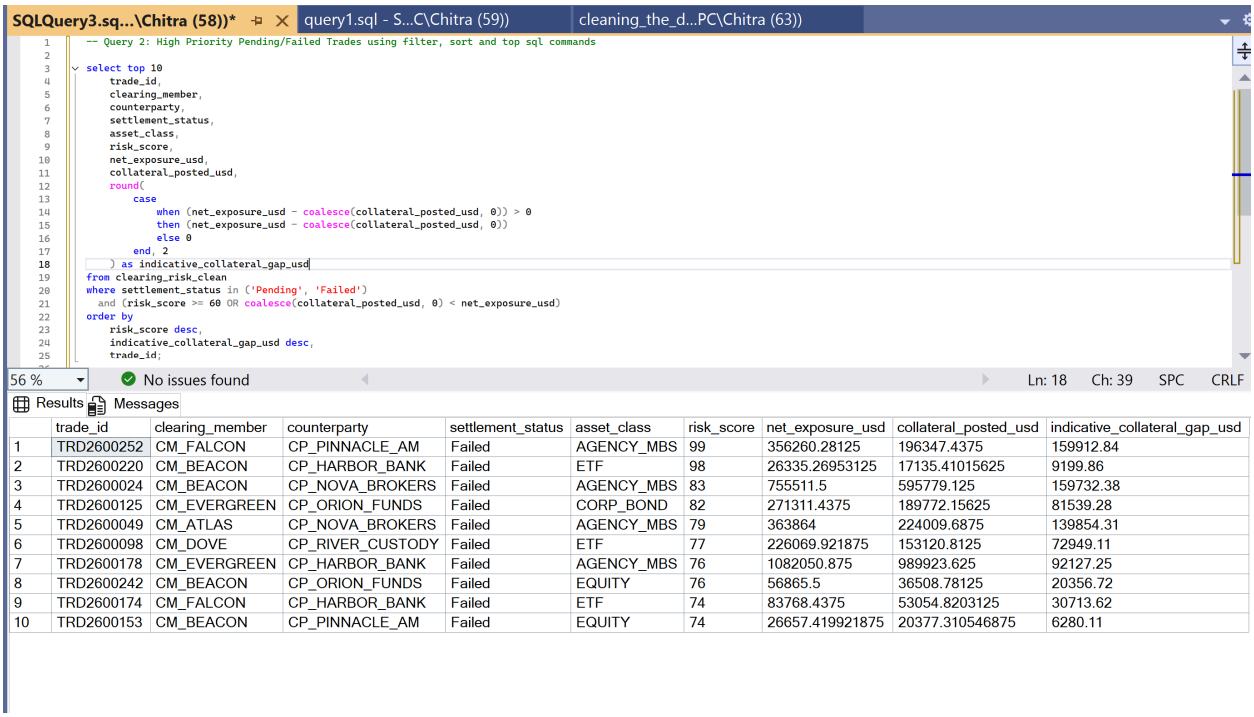
```
SELECT COUNT(*) AS raw_rows FROM clearing_risk_dataset;
SELECT COUNT(*) AS deduplicated_trades FROM clearing_risk_clean;
```

Query 1: Detect duplicated trade identifiers and blank collateral records and missing settlement dates



select
count(*) *as raw_rows,*
count(distinct trade_id) as unique_trade_ids,
*count(**) - count(distinct trade_id) as duplicate_trade_rows,
count(case when collateral_posted_usd is null then 1 end) AS
missing_collateral_rows,
count(case when actual_settlement_date is null then 1 end) AS
missing_actual_settlement_dates
from clearing_risk_dataset;

Query 2: High Priority Pending/Failed Trades using filter, sort and top sql commands



```
-- Query 2: High Priority Pending/Failed Trades using filter, sort and top sql commands

select top 10
    trade_id,
    clearing_member,
    counterparty,
    settlement_status,
    asset_class,
    risk_score,
    net_exposure_usd,
    collateral_posted_usd,
    round(
        case
            when (net_exposure_usd - coalesce(collateral_posted_usd, 0)) > 0
            then (net_exposure_usd - coalesce(collateral_posted_usd, 0))
            else 0
        end, 2
    ) as indicative_collateral_gap_usd
from clearing_risk_clean
where settlement_status in ('Pending', 'Failed')
  and (risk_score >= 60 OR coalesce(collateral_posted_usd, 0) < net_exposure_usd)
order by
    risk_score desc,
    indicative_collateral_gap_usd desc,
    trade_id;
```

No issues found — Ln: 18 Ch: 39 SPC CRLF

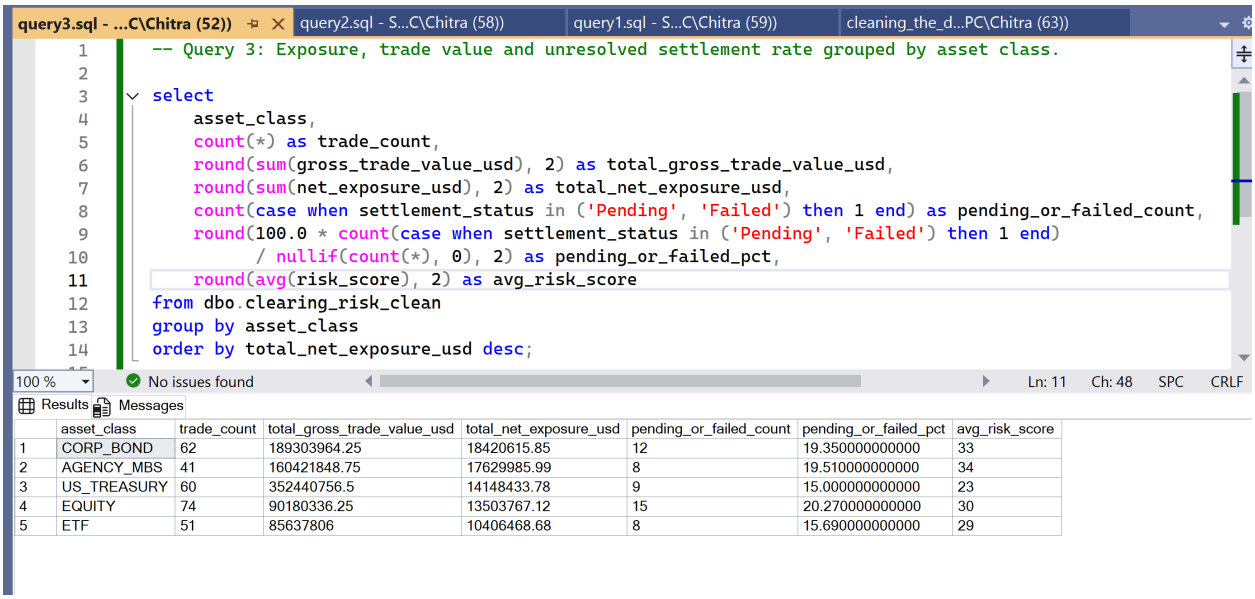
Results | Messages

	trade_id	clearing_member	counterparty	settlement_status	asset_class	risk_score	net_exposure_usd	collateral_posted_usd	indicative_collateral_gap_usd
1	TRD2600252	CM_FALCON	CP_PINNACLE_AM	Failed	AGENCY_MBS	99	356260.28125	196347.4375	159912.84
2	TRD2600220	CM_BEACON	CP_HARBOR_BANK	Failed	ETF	98	26335.26953125	17135.41015625	9199.86
3	TRD2600024	CM_BEACON	CP_NOVA_BROKERS	Failed	AGENCY_MBS	83	755511.5	595779.125	159732.38
4	TRD2600125	CM_EVERGREEN	CP_ORION_FUNDS	Failed	CORP_BOND	82	271311.4375	189772.15625	81539.28
5	TRD2600049	CM_ATLAS	CP_NOVA_BROKERS	Failed	AGENCY_MBS	79	363864	224009.6875	139854.31
6	TRD2600098	CM_DOVE	CP_RIVER_CUSTODY	Failed	ETF	77	226069.921875	153120.8125	72949.11
7	TRD2600178	CM_EVERGREEN	CP_HARBOR_BANK	Failed	AGENCY_MBS	76	1082050.875	989923.625	92127.25
8	TRD2600242	CM_BEACON	CP_ORION_FUNDS	Failed	EQUITY	76	56865.5	36508.78125	20356.72
9	TRD2600174	CM_FALCON	CP_HARBOR_BANK	Failed	ETF	74	83768.4375	53054.8203125	30713.62
10	TRD2600153	CM_BEACON	CP_PINNACLE_AM	Failed	EQUITY	74	26657.419921875	20377.310546875	6280.11

select top 10
trade_id,
clearing_member,
counterparty,
settlement_status,
asset_class,
risk_score,
net_exposure_usd,
collateral_posted_usd,
round(
case
when (net_exposure_usd - coalesce(collateral_posted_usd, 0)) > 0
then (net_exposure_usd - coalesce(collateral_posted_usd, 0))
else 0
end, 2
) as indicative_collateral_gap_usd
from clearing_risk_clean
where settlement_status in ('Pending', 'Failed')
and (risk_score >= 60 OR coalesce(collateral_posted_usd, 0) < net_exposure_usd)
order by

risk_score desc,
indicative_collateral_gap_usd desc,
trade_id;

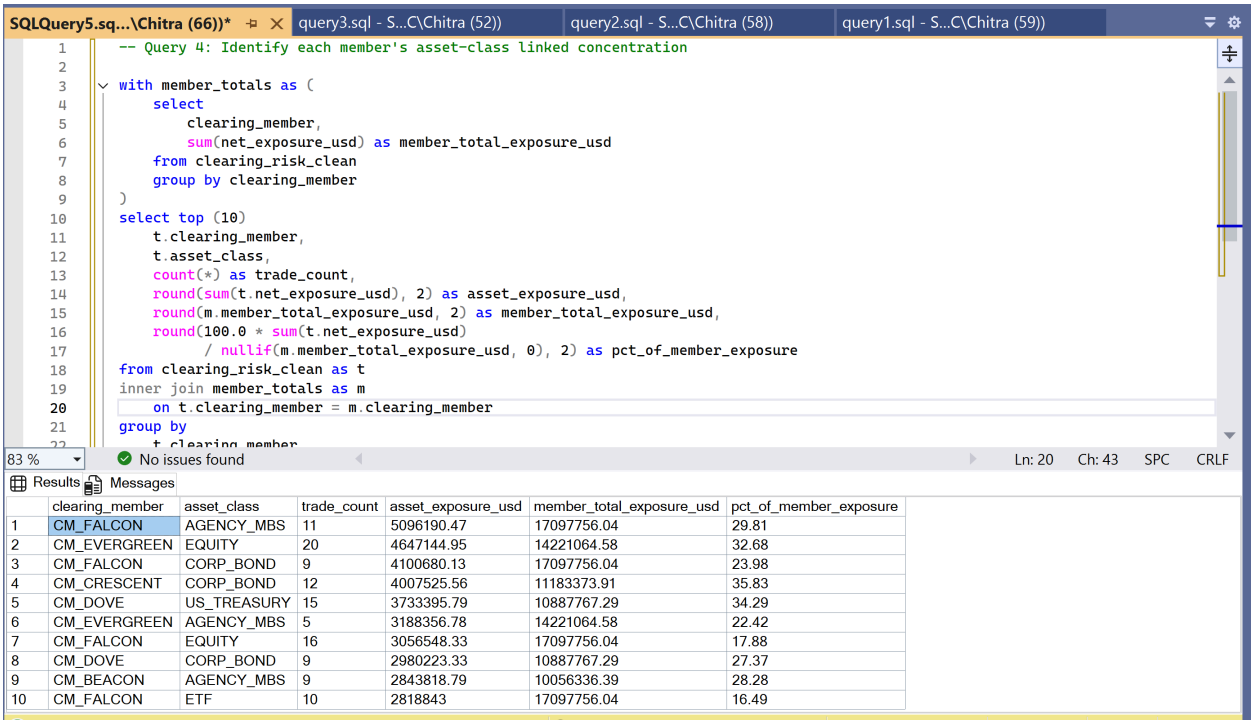
Query 3: Exposure, trade value and unresolved settlement rate grouped by asset class.



	asset_class	trade_count	total_gross_trade_value_usd	total_net_exposure_usd	pending_or_failed_count	pending_or_failed_pct	avg_risk_score
1	CORP_BOND	62	189303964.25	18420615.85	12	19.350000000000	33
2	AGENCY_MBS	41	160421848.75	17629985.99	8	19.510000000000	34
3	US_TREASURY	60	352440756.5	14148433.78	9	15.000000000000	23
4	EQUITY	74	90180336.25	13503767.12	15	20.270000000000	30
5	ETF	51	85637806	10406468.68	8	15.690000000000	29

select
asset_class,
count*() as trade_count,*
round(sum(gross_trade_value_usd), 2) as total_gross_trade_value_usd,
round(sum(net_exposure_usd), 2) as total_net_exposure_usd,
count(case when settlement_status in ('Pending', 'Failed') then 1 end) as pending_or_failed_count,
*round(100.0 * count(case when settlement_status in ('Pending', 'Failed') then 1 end)*
/ nullif(count(), 0), 2) as pending_or_failed_pct,
round(avg(risk_score), 2) as avg_risk_score
from dbo.clearing_risk_clean
group by asset_class
order by total_net_exposure_usd desc;

Query 4: Identify each member's asset-class linked concentration

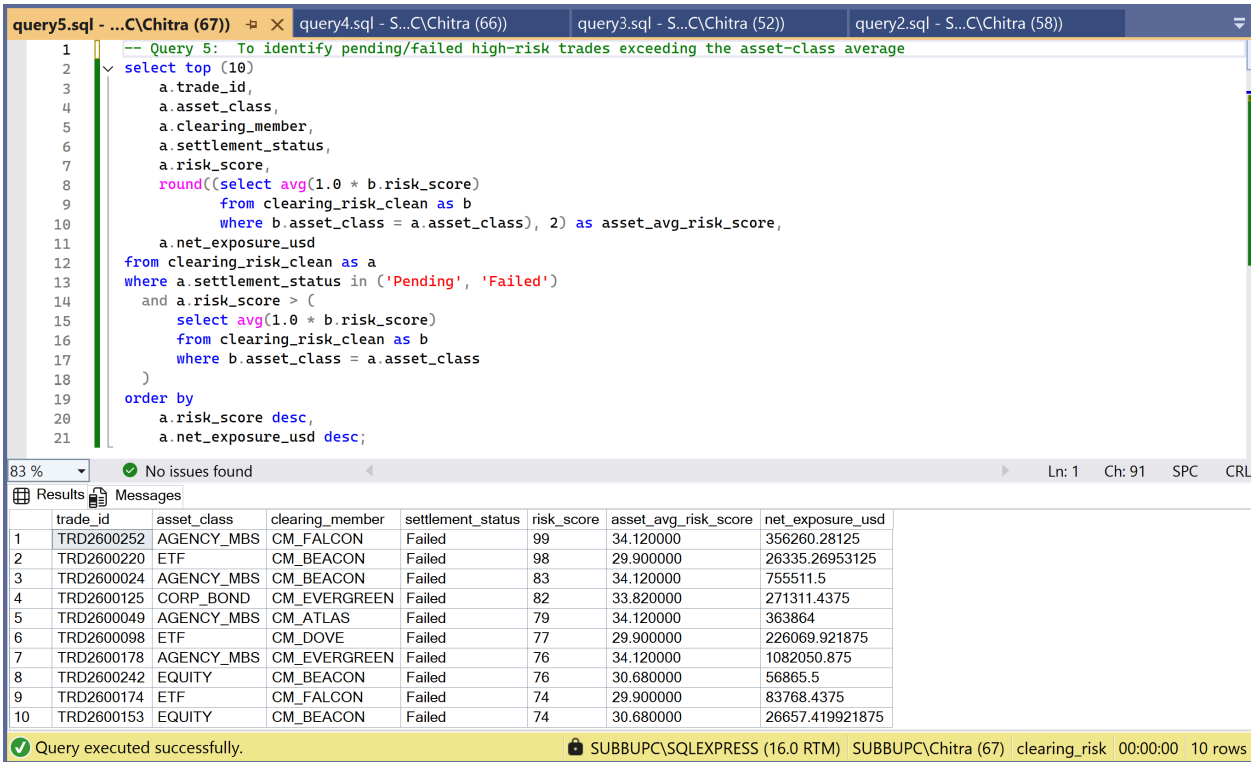


	clearing_member	asset_class	trade_count	asset_exposure_usd	member_total_exposure_usd	pct_of_member_exposure
1	CM_FALCON	AGENCY_MBS	11	5096190.47	17097756.04	29.81
2	CM_EVERGREEN	EQUITY	20	4647144.95	14221064.58	32.68
3	CM_FALCON	CORP_BOND	9	4100680.13	17097756.04	23.98
4	CM_CRESCENT	CORP_BOND	12	4007525.56	11183373.91	35.83
5	CM_DOVE	US_TREASURY	15	3733395.79	10887767.29	34.29
6	CM_EVERGREEN	AGENCY_MBS	5	3188356.78	14221064.58	22.42
7	CM_FALCON	EQUITY	16	3056548.33	17097756.04	17.88
8	CM_DOVE	CORP_BOND	9	2980223.33	10887767.29	27.37
9	CM_BEACON	AGENCY_MBS	9	2843818.79	10056336.39	28.28
10	CM_FALCON	ETF	10	2818843	17097756.04	16.49

```
with member_totals as (
select
clearing_member,
sum(net_exposure_usd) as member_total_exposure_usd
from clearing_risk_clean
group by clearing_member
)
select top (10)
t.clearing_member,
t.asset_class,
count(*) as trade_count,
round(sum(t.net_exposure_usd), 2) as asset_exposure_usd,
round(m.member_total_exposure_usd, 2) as member_total_exposure_usd,
round(100.0 * sum(t.net_exposure_usd)
/ nullif(m.member_total_exposure_usd, 0), 2) as pct_of_member_exposure
from clearing_risk_clean as t
```

inner join member_totals as m
on t.clearing_member = m.clearing_member
group by
t.clearing_member,
t.asset_class,
m.member_total_exposure_usd
order by
asset_exposure_usd desc;

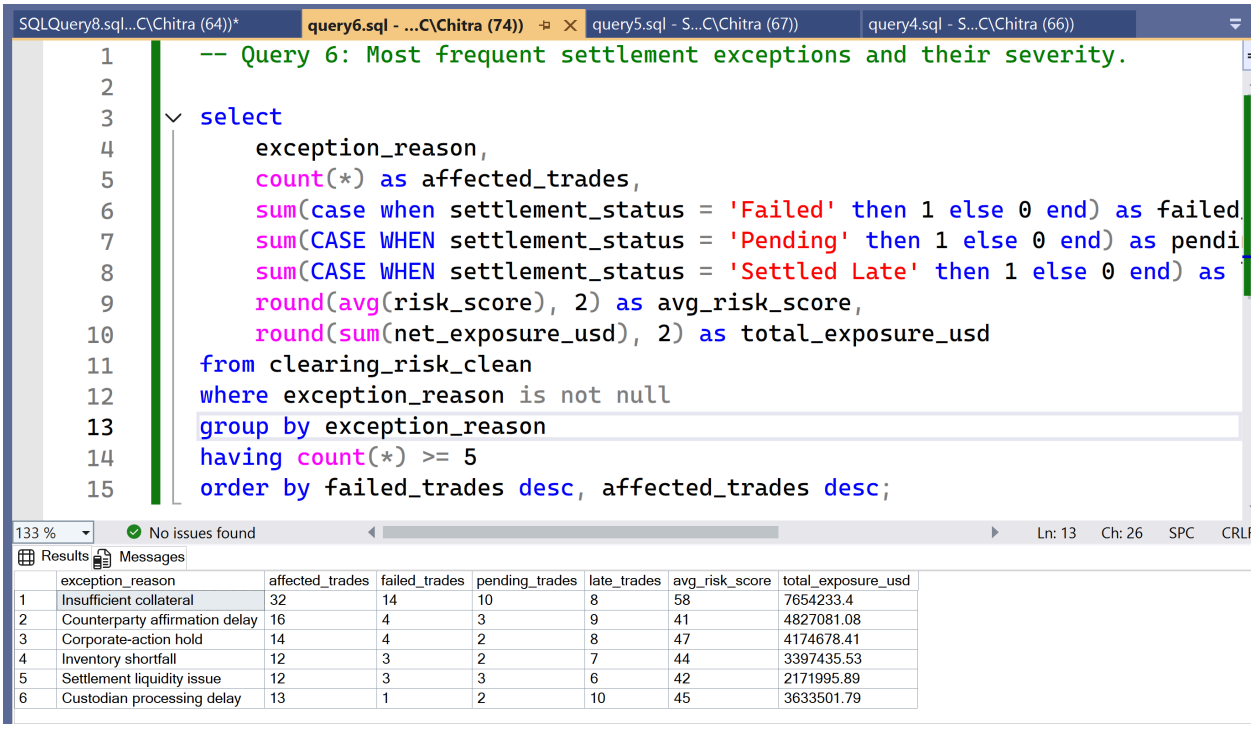
Query 5: To identify pending/failed high-risk trades exceeding the asset-class average



select top (10)
a.trade_id,
a.asset_class,
a.clearing_member,
a.settlement_status,
a.risk_score,

```
round((select avg(1.0 * b.risk_score)
from clearing_risk_clean as b
where b.asset_class = a.asset_class), 2) as asset_avg_risk_score,
a.net_exposure_usd
from clearing_risk_clean as a
where a.settlement_status in ('Pending', 'Failed')
and a.risk_score > (
select avg(1.0 * b.risk_score)
from clearing_risk_clean as b
where b.asset_class = a.asset_class
)
order by
a.risk_score desc,
a.net_exposure_usd desc;
```

Query 6: Most frequent settlement exceptions and their severity.

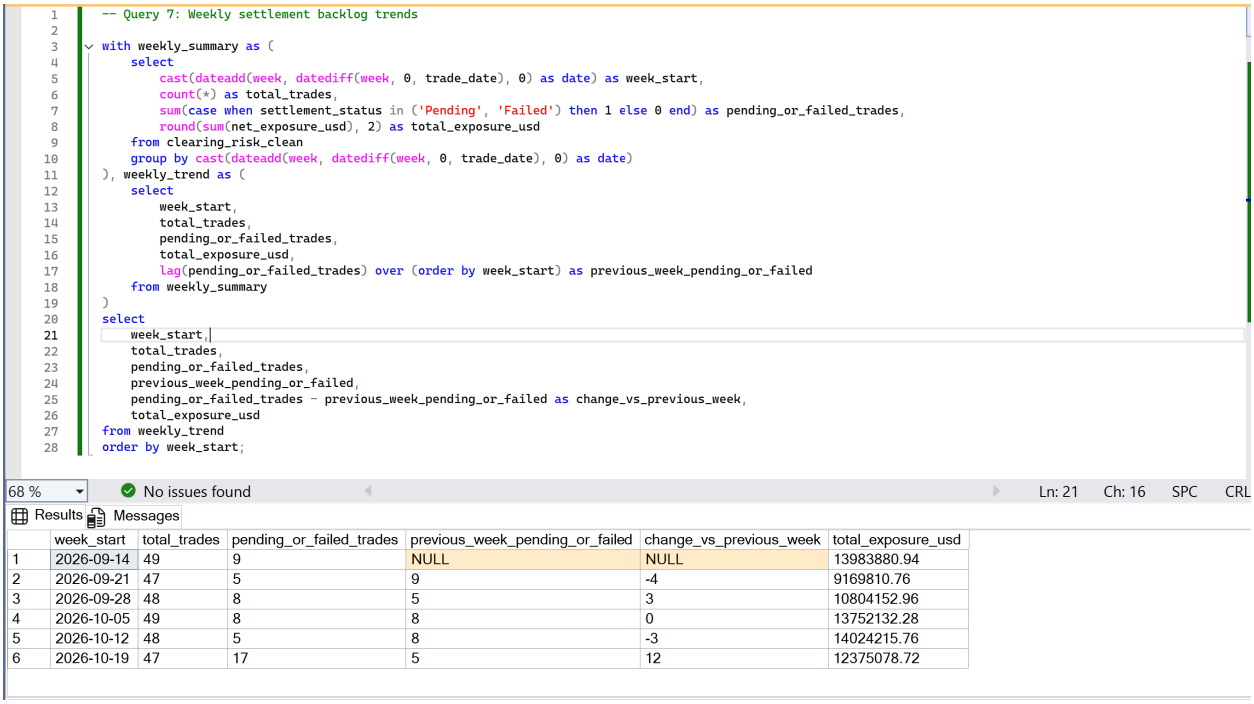


	exception_reason	affected_trades	failed_trades	pending_trades	late_trades	avg_risk_score	total_exposure_usd
1	Insufficient collateral	32	14	10	8	58	7654233.4
2	Counterparty affirmation delay	16	4	3	9	41	4827081.08
3	Corporate-action hold	14	4	2	8	47	4174678.41
4	Inventory shortfall	12	3	2	7	44	3397435.53
5	Settlement liquidity issue	12	3	3	6	42	2171995.89
6	Custodian processing delay	13	1	2	10	45	3633501.79

```
select
exception_reason,
```

count() *as affected_trades,*
sum(case when settlement_status = 'Failed' then 1 else 0 end) as failed_trades,
sum(CASE WHEN settlement_status = 'Pending' then 1 else 0 end) as pending_trades,
sum(CASE WHEN settlement_status = 'Settled Late' then 1 else 0 end) as late_trades,
round(avg(risk_score), 2) as avg_risk_score,
round(sum(net_exposure_usd), 2) as total_exposure_usd
from clearing_risk_clean
where exception_reason is not null
group by exception_reason
having count() >= 5
order by failed_trades desc, affected_trades desc;

Query 7: Weekly settlement backlog trends



```sql
-- Query 7: Weekly settlement backlog trends

with weekly_summary as (
    select
        cast(dateadd(week, datediff(week, 0, trade_date), 0) as date) as week_start,
        count(*) as total_trades,
        sum(case when settlement_status in ('Pending', 'Failed') then 1 else 0 end) as pending_or_failed_trades,
        round(sum(net_exposure_usd), 2) as total_exposure_usd
    from clearing_risk_clean
    group by cast(dateadd(week, datediff(week, 0, trade_date), 0) as date)
), weekly_trend as (
    select
        week_start,
        total_trades,
        pending_or_failed_trades,
        total_exposure_usd,
        lag(pending_or_failed_trades) over (order by week_start) as previous_week_pending_or_failed
    from weekly_summary
)
select
    week_start,
    total_trades,
    pending_or_failed_trades,
    previous_week_pending_or_failed,
    pending_or_failed_trades - previous_week_pending_or_failed as change_vs_previous_week,
    total_exposure_usd
from weekly_trend
order by week_start;
```

68 % No issues found Ln: 21 Ch: 16 SPC CRL

Results Messages

	week_start	total_trades	pending_or_failed_trades	previous_week_pending_or_failed	change_vs_previous_week	total_exposure_usd
1	2026-09-14	49	9	NULL	NULL	13983880.94
2	2026-09-21	47	5	9	-4	9169810.76
3	2026-09-28	48	8	5	3	10804152.96
4	2026-10-05	49	8	8	0	13752132.28
5	2026-10-12	48	5	8	-3	14024215.76
6	2026-10-19	47	17	5	12	12375078.72

with weekly_summary as (
select
cast(dateadd(week, datediff(week, 0, trade_date), 0) as date) as week_start,

```
count(*) as total_trades,
sum(case when settlement_status in ('Pending', 'Failed') then 1 else 0 end) as
pending_or_failed_trades,
round(sum(net_exposure_usd), 2) as total_exposure_usd
from clearing_risk_clean
group by cast(dateadd(week, datediff(week, 0, trade_date), 0) as date)
), weekly_trend as (
select
week_start,
total_trades,
pending_or_failed_trades,
total_exposure_usd,
lag(pending_or_failed_trades) over (order by week_start) as
previous_week_pending_or_failed
from weekly_summary
)
select
week_start,
total_trades,
pending_or_failed_trades,
previous_week_pending_or_failed,
pending_or_failed_trades - previous_week_pending_or_failed as
change_vs_previous_week,
total_exposure_usd
from weekly_trend
order by week_start;
```